

Yes—this can be a strong HRI paper. The correct primary track is still Systems because the contribution is the integration of offline speech, adaptive model routing, personalized memory, and physical soft gestures into one embodied interaction system. HRI explicitly places integrated hardware/software behavior in Systems; a user study is not mandatory when the system evaluation supports the claims. [HRI Systems track](#)

Recommended paper concept

Working title:

An Offline Personalized Conversational Robot with Compute-Aware Model Routing and Soft Embodied Gestures

The central research question should be:

Can an embodied robot provide private, personalized conversation entirely on edge hardware while dynamically selecting model size and personal-memory retrieval to reduce latency and computation?

Proposed architecture



The router makes two separate decisions:

1. Memory decision: RAG or no RAG?
2. Compute decision: small model or larger model?

That produces four possible routes:

Route	Example
Small model, no RAG	“Hello,” “Thank you,” simple social exchanges
Small model with RAG	“What kind of tea do I prefer?”
Large model, no RAG	Complex general reasoning or open conversation
Large model with RAG	“Considering what happened this week, help me plan tomorrow.”

This is clearer than calling the method “hierarchical speculative decoding.” Use the term cascaded model routing or compute-aware routing. Speculative decoding was central to the previous work and should not be reused as the claimed novelty.

Recommended model stack

Component	Recommended starting point	Reason
Speech-to-text	Faster-Whisper small.en, quantized	Established offline ASR with lower memory usage; benchmark base.en if necessary. Faster-Whisper
Small router/model	Qwen3-0.6B, non-thinking mode	Small, Apache-2.0 licensed, and supports structured instruction following. Qwen3-0.6B
Larger generator	Qwen3-4B, 4-bit quantized	Stronger dialogue model; test actual memory and latency before committing. Qwen3-4B
Embeddings	BGE-small-en-v1.5	Compact local embedding model suitable for personal-memory retrieval. Model card
Retrieval store	FAISS plus SQLite metadata	FAISS retrieves vectors; SQLite records date, consent, sensitivity, and source.
Text-to-speech	Piper for speed; Kokoro as quality alternative	Both operate locally; choose through latency testing.
Gesture controller	Separate microcontroller with predefined gesture IDs	Prevents unrestricted LLM output from directly controlling actuators.

Hardware choice

Use one main platform, not both, initially.

- Jetson Orin Nano Super 8GB: better research story because routing and resource efficiency genuinely matter. NVIDIA lists 8GB memory and a 7–25W power range. [Official specifications](#)
- Jetson AGX Thor: easier deployment because it has 128GB memory, but its large compute capacity makes the “saving scarce edge resources” argument less compelling. It can serve as an upper-bound comparison if time permits. [Jetson Thor specifications](#)

If by “Jetson Nano” you mean the original 4GB Jetson Nano rather than Orin Nano, a resident 0.6B-plus-4B pipeline is unlikely to be practical. You would need a much smaller generator or sequential model loading.

Personal memory dataset

The one-week memory should be implemented now, not listed entirely as future work. Future work can be multi-week, multi-user, and automatically learned memory.

Each memory record should contain:

```
Timestamp
Memory text
Memory type: event / preference / routine / fact
Sensitivity level
Consent status
Source
Expiration or deletion status
```

Example week:

- Monday: user started a robotics project
- Tuesday: user prefers morning meetings
- Wednesday: user dislikes overly long answers
- Thursday: user has an appointment next Monday
- Friday: user changed tea preference
- Saturday: user completed a project milestone
- Sunday: user plans to travel

Queries should test:

- Direct facts
- Preferences
- Temporal relationships
- Updated or contradictory information
- Recent versus older events
- Questions for which no memory exists
- Requests involving sensitive or non-consented information

Quantitative benchmark

One real person over one week is sufficient for a demonstration but not for generalizable evaluation. Use:

1. A public conversational-memory benchmark for quantitative results.
2. Your seven-day personal-memory case study for the physical demonstration.

LoCoMo is a relevant benchmark: its conversations average approximately 600 turns and span up to 32 sessions, with question-answering and temporal-memory evaluation. You can evaluate on a smaller, clearly documented subset for the deadline. [LoCoMo paper](#)

If you use actual personal information from a person, obtain explicit consent and confirm the appropriate institutional ethics process before collecting it. Otherwise, use a fictional seven-day persona and describe it as a controlled benchmark—not human-participant data.

Soft-robotic gestures

Use four or five safe, predefined gestures:

- Listening posture
- Gentle nod
- Acknowledgement movement
- Thinking or processing gesture
- Neutral reset

The language model should output a constrained command such as:

```
{
  "speech": "You mentioned that you prefer meetings in the morning.",
  "gesture_id": "ACKNOWLEDGE",
  "memory_used": ["day_2_preference_03"]
}
```

The gesture controller validates the command and executes only known gesture IDs.

Be precise about terminology: if the compliance comes from pneumatic, tendon-driven, elastomeric, or other soft actuation, call it soft robotics. If it is a normal servo under a soft covering, call it compliant or expressive robotic motion.

Without a user study, you may claim that gestures were synchronized and reliably executed. You cannot claim they improved empathy, engagement, trust, or naturalness.

Required experiments

1. Router evaluation

Create an independently annotated test set containing simple, complex, personal-memory, and ordinary queries.

Report:

- RAG/no-RAG decision accuracy

- Small/large model decision accuracy
- Percentage of queries handled by the small model
- Incorrect escalations and incorrect non-escalations
- Compute saved relative to always using the large model

2. RAG evaluation

Report:

- Recall@k
- Mean reciprocal rank
- Answer accuracy
- Temporal-question accuracy
- Updated-preference accuracy
- Correct abstention when information is absent
- Hallucination rate
- Whether answers cite the correct local memory record

3. Model-routing baselines

Compare:

1. Always small model, no RAG
2. Always large model, no RAG
3. Always large model with full memory/RAG
4. Your adaptive router

Measure:

- End-to-end latency
- Answer correctness
- Memory-answer accuracy
- Peak RAM/VRAM
- Power or energy per interaction
- Large-model invocation rate

4. Physical-system evaluation

Report:

- Speech-recognition latency
- Routing latency
- Retrieval latency
- Generation latency
- TTS latency
- Total response latency: median and 95th percentile
- Gesture-command success rate
- Gesture actuation latency
- Speech-gesture onset difference
- Reliability over repeated actuation cycles

- Verification that interaction works without network access

Track decision

Track	Fit
Systems	Best fit. The contribution is the complete offline personalized embodied system.
Technical	Appropriate only if you introduce and rigorously evaluate a genuinely new routing algorithm as the paper's primary contribution.
Design	Appropriate only if the soft-robot mechanism or gesture-design process becomes the main contribution.
User Studies	Appropriate only if the principal contribution comes from a properly approved human study.

Submit to:

Primary: Systems
Optional second choice: Technical

Your defensible contributions

The paper can claim four contributions:

1. An independently developed offline conversational robot combining local speech recognition, generation, retrieval, TTS, and soft embodied gestures.
2. A two-dimensional router that independently selects model size and whether personal-memory retrieval is needed.
3. A local, temporally structured personal-memory system evaluated on conversational-memory questions and a seven-day case study.
4. A physical-system evaluation covering accuracy, retrieval, latency, resource consumption, offline operation, and speech-gesture synchronization.

Voice prosody, camera-based perception, multimodal emotion estimation, multi-user memory, and long-term field deployment should remain future work. This scope is achievable and sufficiently distinct from KATZE-AURA, provided every implementation, dataset, robot design, experiment, and result is new.